O-D-1380A
December 1, 1972
SUPERSEDING
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July 12, 1968

FEDERAL SPECIFICATION

DRY CHEMICAL, FIRE EXTINGUISHING, MULTIPURPOSE PHOSPHATE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE

1.1 This specification covers one type of dry chemical fire extinguishing powder composed mainly of ammonium dihydrogen phosphate.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-378	-	Plastic Sheet and Strip, Thin Gauge, Polyolefin.
RR-S-366	-	Sieve Test.
PPP-D-705	-	Drum, Metal Shipping, Steel (Over 12 and Under 55 Gallon).
PPP-D-729	-	Drum, Metal, 55 Gallons (For Shipment of Noncorrosive Material).
PPP-P-704	-	Pails, Shipping, Steel (1 Through 12 Gallons).

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

FSC 4210

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Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-147 - Palletized and Containerized Unit Loads 40" X 48" Pallet, Skids, Runner, or Pallet-Type Base.

(Copies of Military Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standard:

D-5 - Penetration of Bituminous Materials.

(Application for copies of publications should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc. Agent:

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 S. Riverside Plaza, Chicago, Illinois 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Chemical and physical requirements. The dry chemical powder shall conform to the requirements of table I. Inert water insoluble ingredients and active water soluble ingredients added to increase the overall fire fighting effectiveness of the composition shall be nontoxic and shall not give off toxic vapors when subjected to intense heat.

TABLE I. Chemical and physical requirements

Requirements	Value	Test paragraph
Ammonium dihydrogen phosphate ($NH_4H_2PO_4$) content percent by weight, minimum.	75.0	4.4.1
Inert water insoluble ingredients, percent by weight, maximum.	5.0	4.4.1
Active water soluble ingredients, percent by weight, maximum.	20.0	4.4.1
Moisture content, percent by weight, maximum.	0.20	4.4.2
Hygroscopicity, percent gain in weight, maximum.	1.5	4.4.3
Apparent density, grams per milliliter, minimum.	0.820	4.4.4
Caking and lumping tendency, minimum penetration.	15 mm. or 150 units	4.4.5
Packed density, grams per milliliter, minimum.	1.10	4.4.6
Color identification.	yellow	4.4.7

3.1.1 Particle size. The particle size distribution of the dry chemical powder shall conform to the requirements of table II.

TABLE II. Particle size requirements

U. S. screen size	Percent retained	
	Minimum	Maximum
40	0.0	0.0
100	0.0	10.0
200	12.0	25.0
325	12.0	25.0
Bottom pan	50.0	70.0

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3.1.2 Sub-screen particle size analysis. The complete sub-screen particle size analysis of the sample shall be obtained, as specified in 4.4.9. This shall be stated in percentages obtained in particle size groups in accordance with table III. The sample shall show a median particle size in the range from 15 microns to 25 microns and a 40 percent minimum with a 60 percent maximum in the range of 20 microns or less.

TABLE III. Parameters for roller analyzer sub-screen particle size analysis

Size, particles separated (microns)	Chamber diameter (inches)	Nozzle diameter (inches)	Air flow liters/ min.	Manometer reading mm. water	Time in minutes
0-10	9	.082	14.0	5.8	75
10-15	4-1/2	.055	7.9	25.5	30
15-20	4-1/2	.082	14.0	5.8	120
20-25	4-1/2	.096	21.9	12.3	120
25-30	2-1/4	.055	7.9	25.5	90
30-35	2-1/4	.070	10.7	3.5	90
35-40	2-1/4	.082	14.0	5.8	--
40-45	1-1/4	.046	5.5	13.5 1/	--
45 and over	Material remaining in sample tube				

1/ Use 0.066-inch diameter capillary in last run.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall consist of the examination of 4.2.2 and the tests of 4.3.

4.2.1 Sampling.

4.2.1.1 Inspection lot. An inspection lot shall consist of five bathes or less when this fulfills the requirements of an order. When more than five batches are required to fill an order, the batches shall be grouped as equally as possible by weight to give not less than three nor more than five lots.

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4.2.1.2 Test samples. Samples from a lot comprised of five batches or less shall consist of a two-pound composite batch sample and shall immediately be placed in a clean, dry, air and water-tight container which shall be sealed, marked, and sent to a testing laboratory satisfactory to the procuring agency concerned. The sample shall be subjected to all the tests, as specified in 4.3. When more than five batches are required to fulfill an order, the batches shall be grouped as equally as possible by weight to give not less than three nor more than five lots. When the lot consists of more than five batches, a sufficient quantity shall be taken from each batch to make a ten-pound composite lot sample. Each lot composite sample shall be prepared for testing as above.

4.2.2 Examination of filled containers. A random sample of filled containers shall be selected from each lot in accordance with MIL-STD-105 at inspection level I, and an acceptable quality level (AQL) of 2.5 percent defective to verify compliance with all stipulations of this specification regarding fill, closure, packaging, packing, marking, and other requirements not involving tests. Containers shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample having one or more defects or under required fill shall be rejected, and if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of MIL-STD-105, the lot represented by the sample shall be rejected.

4.3 Tests. Samples selected in accordance with 4.2.1.2 shall be tested for conformance with all the section 3 requirements. If the test sample fails to pass one or more of these tests, the lot represented shall be rejected.

4.4 Test methods. Unless otherwise specified herein, all tests shall be run in triplicate and the results averaged.

4.4.1 Composition. Standard officially recognized analytical procedures for the analysis of ammonium dihydrogen phosphate may be used and the results reported in $NH_4H_2PO_4$, percent by weight. Inert water insoluble ingredients shall also be reported in terms of percent by weight. The remainder of the composition shall be water soluble and shall make up the total of 100 percent of the composition.

4.4.2 Moisture content. In an atmosphere of 50 percent or less relative humidity, weigh accurately a sample of approximately 50 grams (± 1 mg.) into a tared aluminum dish. Place the dish in a desiccator using H_2SO_4 (A.C.S. reagent grade, 95 percent to 98 percent by weight) as a drying agent. Maintain the closed desiccator and contents at a temperature of 70° (± 5° F.) for 24 hours. At the end of this time period, remove the test sample and weigh accurately. The moisture content of the sample is calculated as follows:

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$$M = \frac{100\ (W_1 - W_2)}{W_1}$$

Where:
M = moisture content, percent.
W_1 = original weight of sample.
W_2 = weight of sample after 24 hours drying.

4.4.3 Hygroscopicity. Using an aluminum moisture dish as a sample container, overfill with the material and then strike off with a straight edge. Cap the dish with its cover and weigh accurately. Remove the cover and place the sample in a humidifier with circulating air (as described in paragraph 4.4.5) maintained at 78 percent R.H. and 70° F. (± 5° F.) for a period of 48 hours. At the end of this period, cover the sample, remove and weigh it. Determine the gain in weight and calculate the percentage gain in weight in the sample.

4.4.4 Apparent density. Place 100 grams (± 0.1 gram) of sample in a clean, dry 250 ml. glass-stoppered graduated cylinder, having an inside diameter of approximately 3.5 cm. Stopper the cylinder and rotate slowly end over end for 10 complete cycles. After this agitation, set the cylinder upright on a surface and allow to settle for 3 minutes. Read the volume occupied by the sample and report as apparent density as follows:

$$\text{Apparent density (g. per ml.)} = \frac{100\ \text{g.}}{\text{Volume of sample, ml.}}$$

4.4.5 Caking and lumping tendency. Place 125 gram samples in a nickel crucible-shaped cups (100 ml. capacity), then vibrate for five minutes on a Tyler Ro-Tap or equivalent sieve shaker using a suitable holder. Then place the samples in a moving air stream humidifier of 78 percent R.H. for a period of 24 hours, after which time they are to be placed in a drying oven of 118° F. for a period of 24 hours. At the conclusion of the second period (total 48 hours), the degree of caking is determined with a Penetrometer conforming to the requirements of ASTM D-5. Penetration is defined as the resistance of the material, expressed as the distance that a standard needle penetrates vertically into the sample under known conditions of load and time. The moving weight of the needle used is 50 grams and the time 5 seconds. An average of six readings (three from each sample) shall be used.

Note: It has been found that the stagnant air conditions found in the usual saturated solution desiccator compartments are not conducive to consistent results and precautions must be taken to insure air circulation if a desiccator is used as the humidifier. A 250 mm. diameter desiccator with a saturated NH_4Cl solution in the lower compartment may be used. During operation, a circulating air supply of five liters per minute shall be presaturated by bubbling through a saturated NH_4Cl solution and then introduced into the desiccator through a center tube with 6 mm. inside diameter terminating 2 cm. above the center hole in the desiccator plate. It is advisable to check the relative humidity of the exit air stram from time to time by some convenient means.

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4.4.6 Packed density. The packed density of a dry chemical is obtained by determining its weight per volume after a period of extreme packing produced by vibration. Equipment necessary for conducting the packed density analysis includes a conveniently shortened 250 ml. graduated glass cylinder (approximately 1-11/32 i.d.) fitted with a No. 8 rubber stopper and a Tyler Ro-Tap sieve shaking machine or equivalent. A 100 gram sample of dry chemical is poured into the graduate which is placed in a convenient holder modified to fit the shaking machine. The graduate is shaken on the Ro-Tap machine for 30 minutes. After the shaking period the volume of the powder is read directly in ml. from the graduated cylinder. An average of three runs on one sample is used in making the packed density calculation which is as follows:

$$\text{Packed density (grams per milliliter)} = \frac{\text{(wt. of sample) (grams)}}{\text{Packed volume of sample (ml.)}}$$

4.4.7 Color identification. The powder sample shall be yellow in color when viewed in bulk in its container.

4.4.8 Particle size. The shaker to be employed in conducting particle-size tests shall operate with a single eccentric circulator motion at 285 (± 10) revolutions per minute and with a tapping action of 150 strokes per minute to obtain dependable sieve analysis. To permit flexible operation, the shaker shall accommodate one to six 8-inch diameter sieves with one pan and cover. Screen sizes used for particle size analysis shall conform to U.S. Standard Screen Sizes as outlined in RR-S-366 and shall be stainless steel, nonsupported.

4.4.8.1 Procedure. In an atmosphere of 50 percent R.H. or less, place an accurately weighed sample of approximately 50 grams in the top screen of a nexted series of 8-inch U.S. Standard screens consisting of a 40-, 100-, 200-, and 325-mesh, with cover and bottom pan. Agitate the screen nest for 5 minutes on a shaker employing eccentric circular motion and tapping action such as is used in the Tyler Ro-Tap machine. Separate the screen next and weigh accurately the amount remaining on each, expressing the results in percentage of the original sample.

Note: Care must be taken to remove all the residue from the screen. Sieves should be cleaned with a camels hair brush to insure complete removal of the fine particles.

4.4.9 Sub-screen particle size analysis test. In order to determine the particle size distribution of the sample in sub-screen size ranges, a Roller air analyzer or equivalent is used (see 6.3). The air to be used in the elutriation process in the instrument must be humidified to 80 percent R.H. (if necessary) in order to eliminate building of static charges when separating the sample. Humidification is accomplished by passing the air through water, then through a saturated solution of ammonium chloride. Pass air through the apparatus for at least 30 minutes before starting the determination. At the end of this time, determine the weight of the thimble to the nearest 0.1 mg.

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on an analytical balance. Place the correct nozzle (see table III) in the system and connect to the sample tube to which 15 grams of the test sample has been added. Connect the sample tube to the proper size settling chamber and turn on the tapper and air flow. Adjust the air flow until the desired manometer reading is obtained. Allow the air to pass through the apparatus for the specified length of time. At the end of this period stop the air flow and tapper, remove the thimble, and determine its weight. The test is continued using the next combination of nozzle, air flow, chamber size, and testing time in order to determine the next particle size. Repeat the procedure by collecting the various particle sizes on a cumulative basis. Discontinue the test following that particle size which results in a cumulative value of at least 50 percent. The combination used for each separation are given in table III.

Calculations:

The percentage of material separated in each fraction is calculated as follows:

$$\text{Percent separated} = \frac{\text{final thimble wt. - initial thimble wt.}}{15} \times 100$$

The percentage of the original sample accounted for is determined by adding the percentages separated in each fraction and the percentage remaining in the sample tube. This value can be used to correct the amounts departed to 100 percent.

$$\text{Corrected percentage} = \frac{\text{percent in fraction}}{\text{Total separated}} \times 100$$

Using the corrected values calculate the cumulative percentage of material less than the upper size limit of a group by adding the percentages in each successive group. The percentage of 20 microns or less is the sum of the percentages in the groups 0 to 10, 10 to 15, and 15 to 20 microns. The data is also used to determine a median particle size for the sample and a range of particle size. The median is calculated using the formula:

$$\text{Median} = \frac{A + 5\ (50-C)}{B-C}$$

A = Lower size value in first size group having cumulative percentage value of 50 or over.
B = Cumulative percentage of group with A as lower size value.
C = Cumulative percentage of group with A as upper size value.

4.5 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected to determine conformance with the applicable requirements of section 5 of this specification.

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5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be levels A or C, as specified (see 6.2).

5.1.1 Level A. The material shall be furnished in the following type containers.

5.1.1.1 Pails. Unless otherwise specified, the material shall be packaged in a single bead, 6-gallon pail with wire bail, handle, and grip conforming to PPP-P-704, type II, class 2, except that a 26-gauge cover with a lever-lock, reusable type ring shall be used in lieu of the lug cover. The lever-lock ring and cover shall be fitted with a suitable type rubber sealing gasket.

5.1.1.2 Drums. Material packaged in containers larger than 6-gallon capacity shall be shipped in metal drums with full removable heads conforming to PPP-D-705 or PPP-D-729, as applicable.

5.1.2 Level C. The pipe shall be packed in a manner that will insure arrival at destination in satisfactory condition and be acceptable to the carrier at lowest rates. Containers and packing shall comply with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Palletization. When specified (see 6.2), filled containers shall be palletized in accordance with the applicable requirements of MIL-STD-147.

5.3 Marking. In addition to any special marking required by the contract or order, marking shall be in accordance with 5.3.1 or 5.3.2, as specified (see 6.2).

5.3.1 Civil agencies. Marking shall be in accordance with Fed. Std. No. 123.

5.3.2 Military agencies. Marking shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use on class A, B, or C fires. This agent is not necessarily compatible with all types of protein-type foam.

6.2 Ordering data. Purchasers should select the preferred options offered herein, and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type of container and selection of applicable level of packing (see 5.1).
(c) Palletization, when required (see 5.2).
(d) Special markings, if required (see 5.3).

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

O-D-1380A: Dry Chemical, Fire Extinguishing, Multipurpose Phosphate

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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